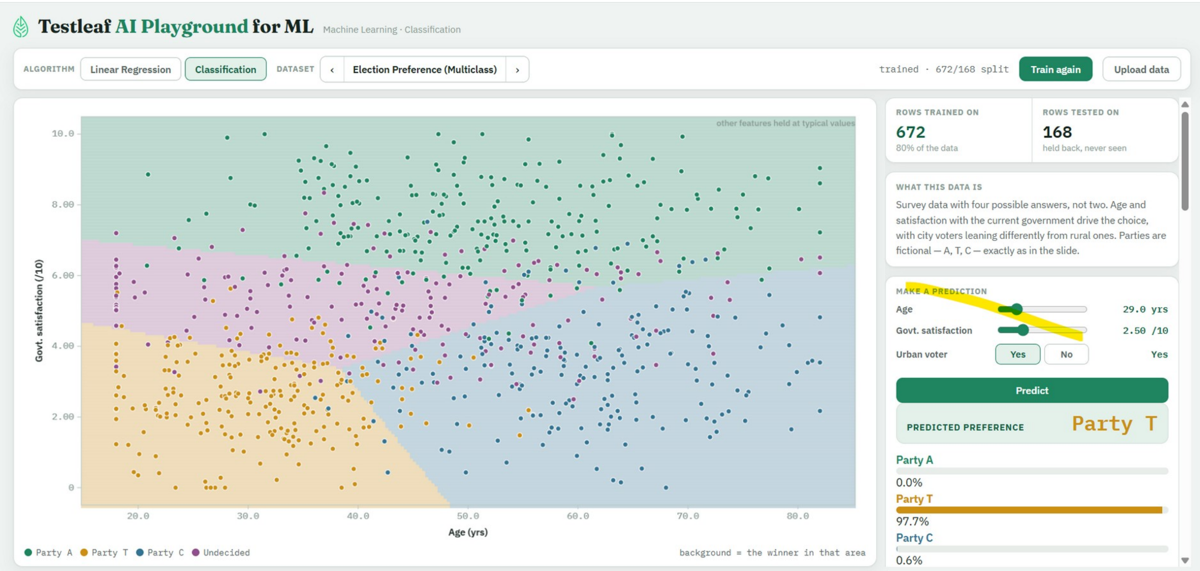
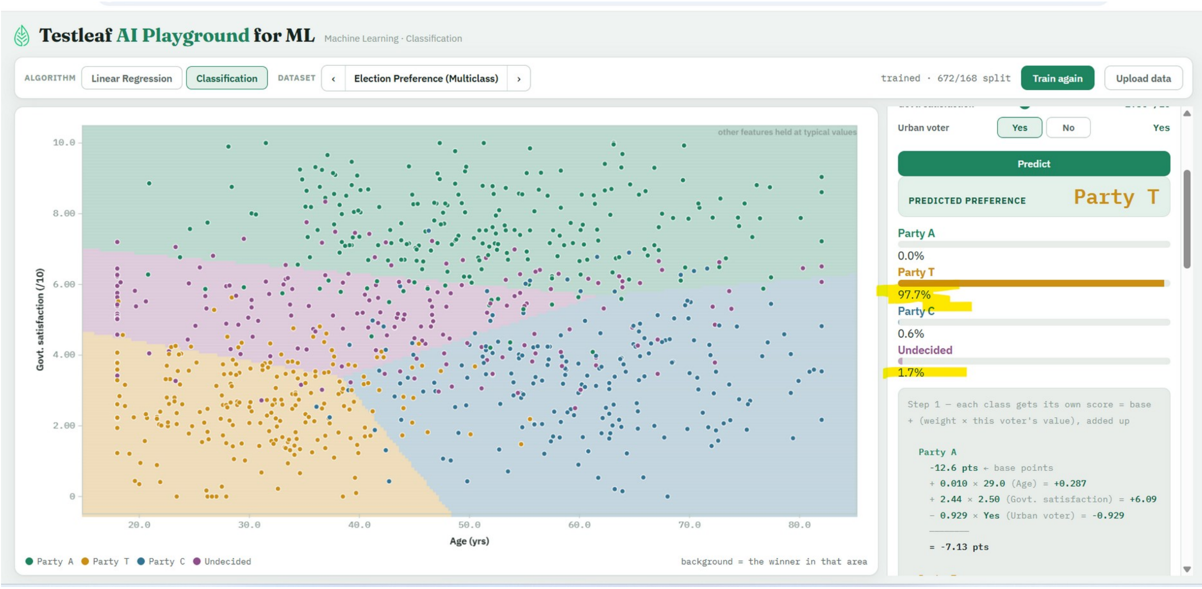
1) Classification Algorithm

1) Age <30 and Govt satisfaction less , urban

2) Age >50 and Govt satisfaction less , non urban validate the percentage.

1.

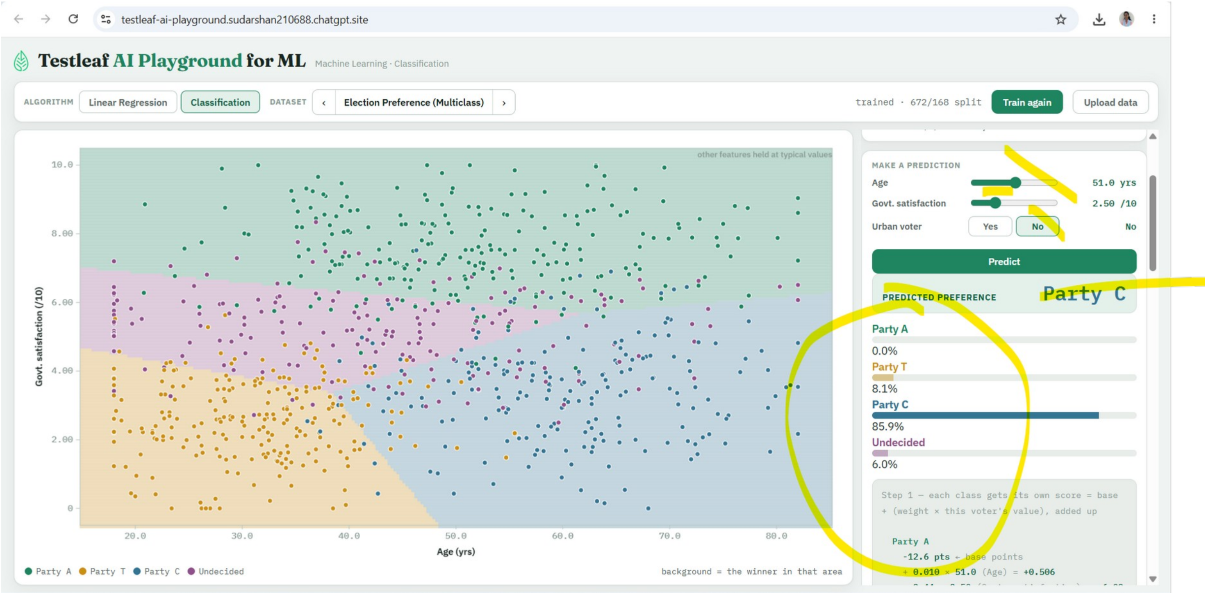



Testleaf AI Playground for ML
Machine Learning · Classification
ALGORITHM
Linear Regression
Classification
DATASET
Election Preference (Multiclass)
trained · 672/168 split
Train again
Upload data
other features held at typical values
Govt. satisfaction (/10)
Age (yrs)
Party A
Party T
Party C
Undecided
background = the winner in that area
Urban voter
Yes
No
Yes
Predict
PREDICTED PREFERENCE
Party T
Party A
0.0%
Party T
97.7%
Party C
0.6%
Undecided
1.7%
Step 1 — each class gets its own score = base + (weight × this voter's value), added up
Party A
-12.6 pts ← base points
+ 0.010 × 29.0 (Age) = +0.287
+ 2.44 × 2.50 (Govt. satisfaction) = +6.09
− 0.929 × Yes (Urban voter) = -0.929
= -7.13 pts


Step 2b — each class's share = its own e^score ÷ that same total (96.7)
Party A 0.001 ÷ 96.7 = 0.0%
Party T 94.4 ÷ 96.7 = 97.7%
Party C 0.593 ÷ 96.7 = 0.6%
Undecided 1.64 ÷ 96.7 = 1.7%
All 4 shares add back up to 100% — that's the whole point of dividing by the same total. e (≈ 2.71828) is just a fixed constant that turns any score positive and stretches the gap between a high score and a low one.

**Party T - 97.7%**

2.




**Party C with 85.9**

Party A

-12.6 pts ← base points

\+ 0.010 × 51.0 (Age) = +0.506

\+ 2.44 × 2.50 (Govt. satisfaction) = +6.09

− 0.929 × No (Urban voter) = +0

────────

= -5.98 pts

Party T

10.2 pts ← base points

− 0.141 × 51.0 (Age) = -7.19

− 1.10 × 2.50 (Govt. satisfaction) = -2.75

\+ 1.23 × No (Urban voter) = +0

────────

= 0.222 pts

Party C

-0.536 pts ← base points

\+ 0.086 × 51.0 (Age) = +4.40

− 0.513 × 2.50 (Govt. satisfaction) =
-1.28

− 1.21 × No (Urban voter) = +0

────────

= 2.58 pts

Undecided

-0.019 pts ← base points

− 0.038 × 51.0 (Age) = -1.94

\+ 0.748 × 2.50 (Govt. satisfaction) =
+1.87

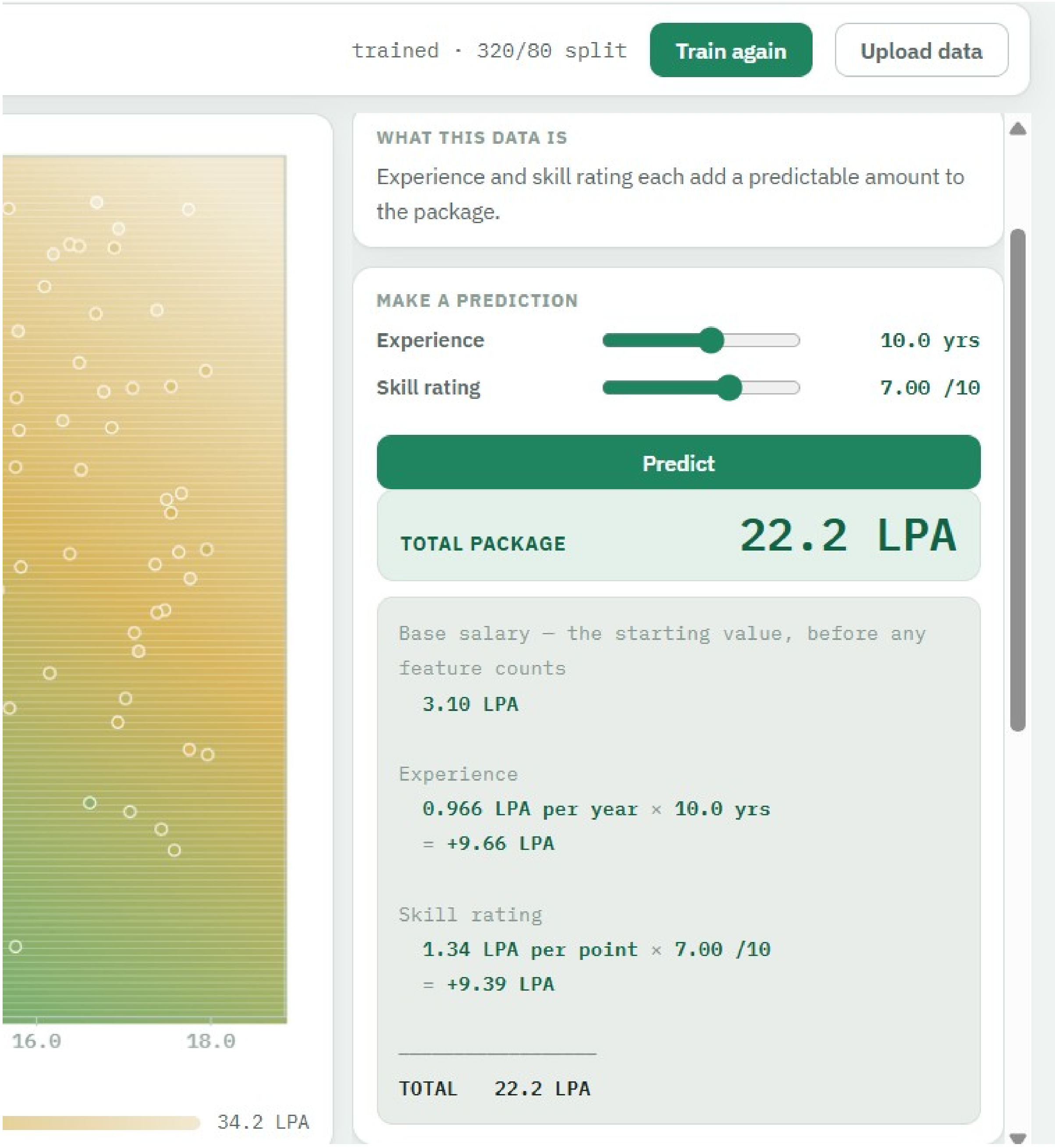
```
    = -0.085 pts
─────────────
highest score wins → Party C
```

```
Step 2a — convert each score to e^score
(always positive)

  Party A     e^-5.98 = 0.003
  Party T     e^0.222 = 1.25
  Party C     e^2.58 = 13.2
  Undecided   e^-0.085 = 0.919
─────────────
Total (add all 4 up)  =  15.4
```

```
Step 2b — each class's share = its own e^score
÷ that same total (15.4)

  Party A     0.003 ÷ 15.4 = 0.0%
  Party T     1.25 ÷ 15.4 = 8.1%
  Party C     13.2 ÷ 15.4 = 85.9%
  Undecided   0.919 ÷ 15.4 = 6.0%
─────────────
All 4 shares add back up to 100% — that's the
whole point of dividing by the same total. e
(≈ 2.71828) is just a fixed constant that
turns any score positive and stretches the gap
between a high score and a low one.
```

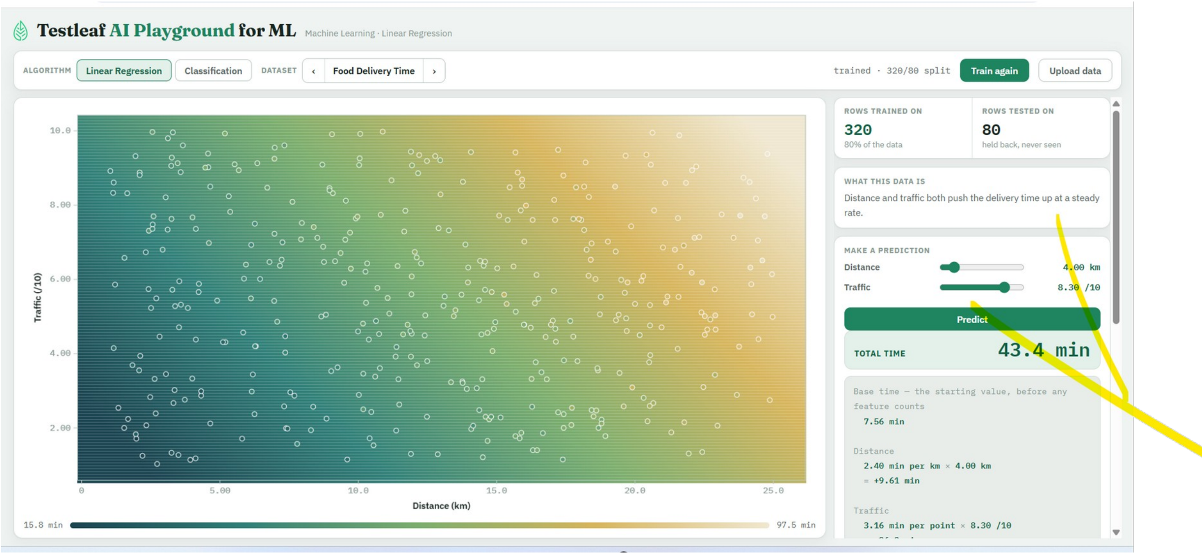
2)Linear regression

1) QA Salary




2) Food Delivery time

Check the Calculation for price



Distance **4.00 km**

Traffic **8.30 /10**

**TOTAL TIME 43.4 min**

7.56 min +

Distance

2.40 min per km × 4.00 km

= +9.61 min +

Traffic

3.16 min per point × 8.30 /10

= +26.2 min

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TOTAL   43.4 min